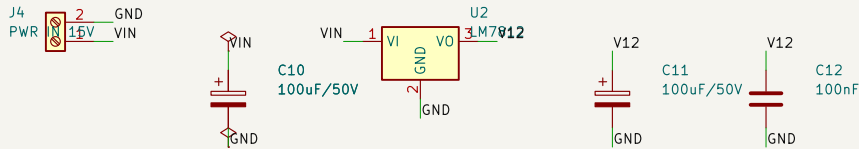


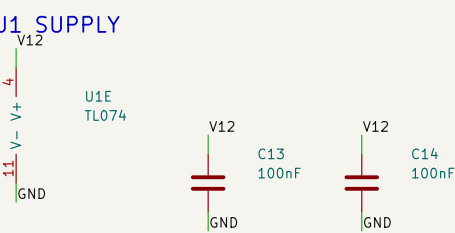
Bose Companion 5 sub crossover -- mono--summing 2nd--order Sallen--Key low--pass.

Single +12 V rail, no negative supply. All AC referenced to VGND = 6 V (A3 buffers the R8/R9 mid--rail divider).

A POWER IN --> +12 V

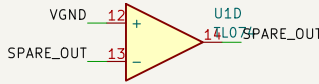


B U1 SUPPLY



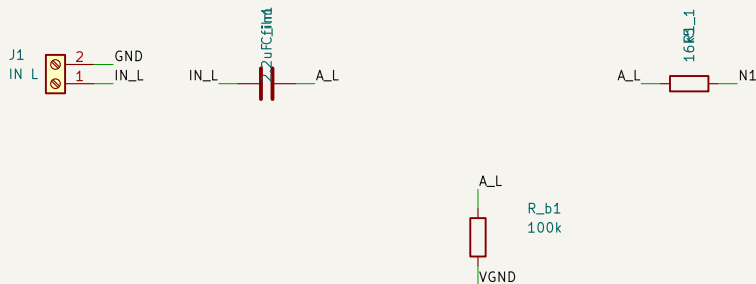
U1 supply: pin 4 = V12, pin 11 = GND. No negative rail.

C VIRTUAL GROUND 6 V



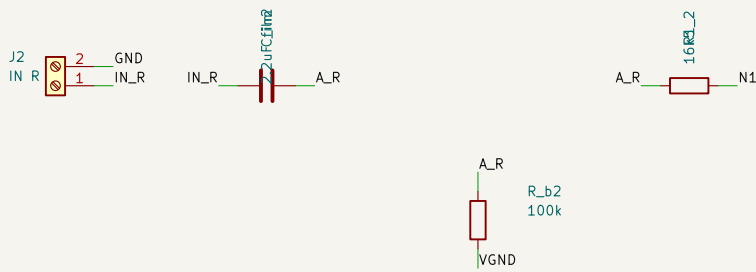
A4 is the spare section: + on VGND, out tied to in--. An unterminated spare oscillates and couples into its neighbours through the shared supply.

D INPUT L

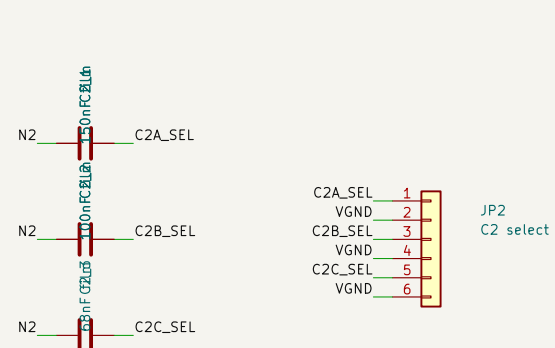
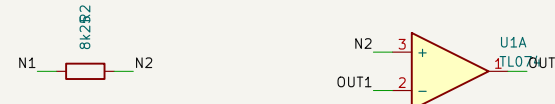
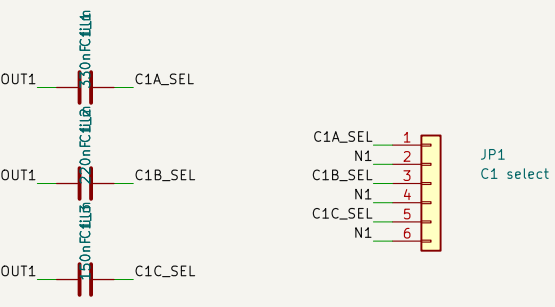


R_b1/R_b2 are the only DC path to A1's + input (C1 blocks DC, C2 is a capacitor). Without them A1 drifts to a rail. They sit after the coupling caps -- 100k at N1 would shift every corner by 8%.

E INPUT R

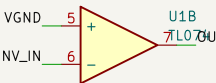
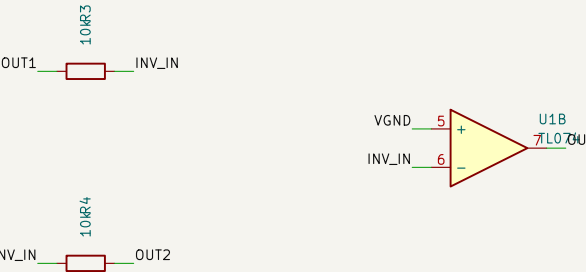


F SALLEN--KEY LOW--PASS

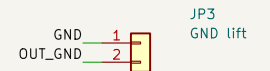
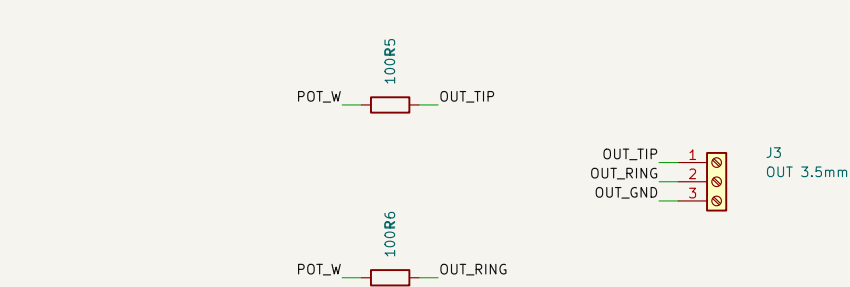
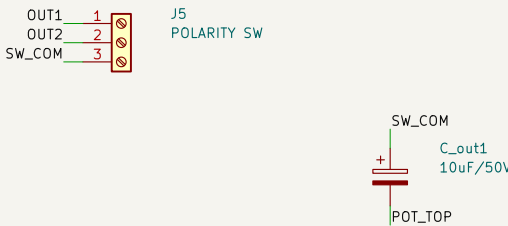


C1 returns to N1 -- the Sallen--Key feedback path. Not N2.
JP1/JP2 set the corner. ONE shunt in each, BOTH in the same position:
1--2 = 87 Hz (330n/150n) 3--4 = 130 Hz (220n/100n) 5--6 = 191 Hz (150n/68n)
A mismatched pair gives the wrong Q, not an obvious fault.

G POLARITY INVERTER



H OUTPUT STAGE



C_out + faces SW_COM: that node sits at 6 V DC, the pot side at 0 V.
J6 pin 3 = GND, not VGND -- C_out has already removed the 6 V bias; referencing the pot to VGND would thump the driver at power--on.
JP3 shorts R7 (ground lift). Normally fitted.

Off--board: level pot (J6), polarity switch (J5), all panel jacks (J1/J2/J3).
Process: single--sided THT, bottom copper only, board outline 100 x 100 mm.
Copper is isolation--milled on the CNC with an 0.8 mm flat end mill; the fiber laser does the silkscreen only. Routing clearance 0.85 mm / track 1.0 mm.
The 0.8 mm tool width is a hard floor on every copper--to--copper gap. Fixed--pitch pads sit below the 0.85 mm target and carry a kicad.dru exception: DIP--14 0.94 mm (passes), 2.54 mm headers and the TD--220 0.84 mm (do not).
Drills: 0.8 mm x52, 1.0 x26, 1.1 x3, 1.2 x4, 1.5 x15.

Reference names vs the handoff document: KiCad treats a reference with no trailing digit as unannotated, so C1a/C1b/C1c --> C1_1/C1_2/C1_3, C2a/C2b/C2c --> C2_1/C2_2/C2_3, R1a/R1b --> R1_1/R1_2, C_out --> C_out1.

Single--sided THT, CNC isolation milling (0.8 mm end mill), laser silkscreen
Mono--summing 2nd--order Sallen--Key low--pass, single +12 V rail

Sheet: /

File: subxo.kicad_sch

Title: Bose Companion 5 sub crossover

Size: A2

Date: 2026--08--13

Rev: A

KiCad E.D.A. 10.0.4

Id: 1/1